Datasets as Medium:

Teaching the Full ML Pipeline from Collection to Synthesis

Learning To Teach: (Re)Designing Creative Tech Pedagogy for the GenAI Era (#LTT2026)

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Helena Sarin, *Leaves of Manifold* (2017). The artist trained a generative AI system on scans of approximately 1000 leaves collected from her own backyard.

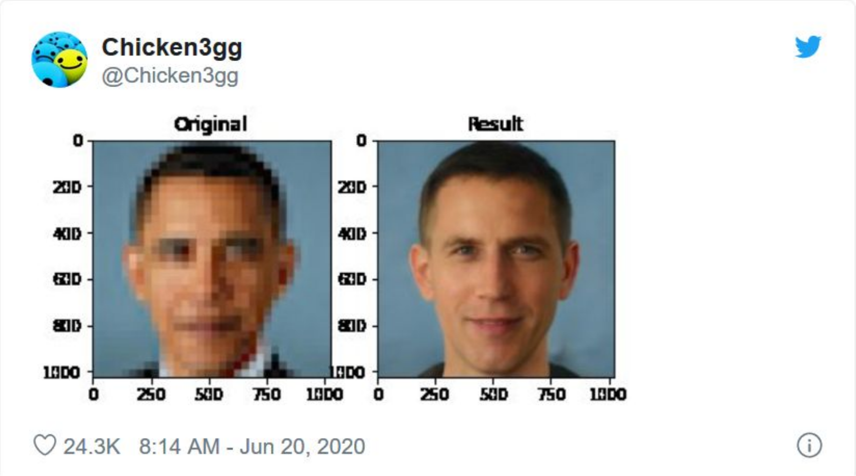
- Datasets are the locus of creativity in ML-based art.
- Datasets are the hardest part in ML-based art.

This assignment is concerned with data sovereignty. We will collect our own training dataset, and train our own model(s) with it.

Along the way, we will use machine learning to understand something new about our data itself.

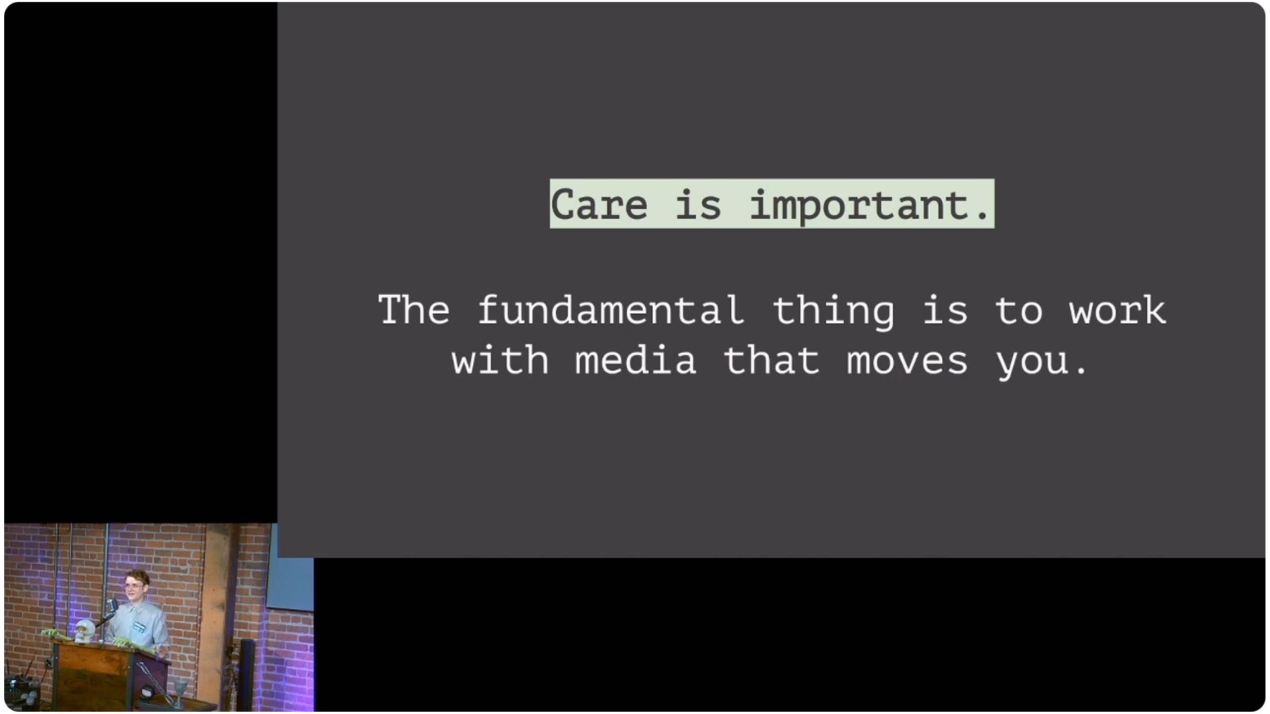
Datasets are biased

- Datasets are made by humans, and reflect their biases.
- Datasets are usually made for business or surveillance purposes (self-driving cars, advertising, face recognition)... not art



This face was upscaled using a model that was trained on data with biases *(Via Derrick Schultz)*

3.1a. Background: Corpora as Medium, Curating as Creative Practice



Everest Pipkin - Corpora as medium: on the work of curating a poetic textual dataset

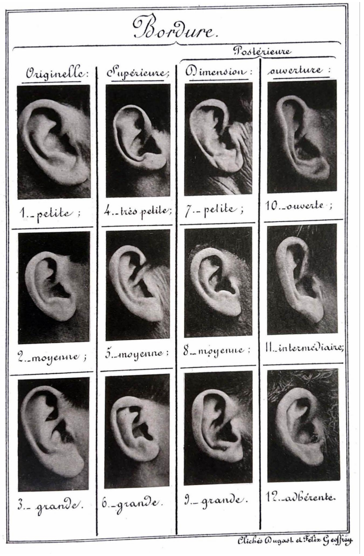
The unit begins with a viewing of Everest Pipkin's 23-minute talk from 2019, *Corpora as medium: on the work of curating a poetic textual dataset,* and a reading of their 2020 article, *On Lacework: watching an entire machine-learning dataset.*

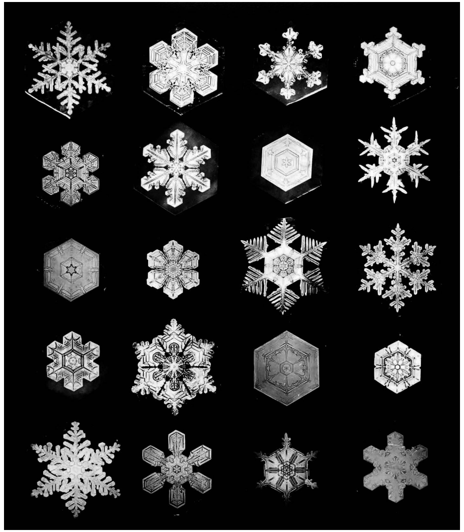
Pipkin's practice has involved the collection, production, and/or filtering of datasets in ways that sometimes involve extreme labor and endurance. During the pandemic, they viewed the entirety of MIT's *Moments In Time* dataset — one million 3-second videos — and selected their favorites.

Pipkin notes that the words "**curation**" and "**care**" come from the same root.

3.1b. Background: Typologies and Small Multiples

Students consider [typologies](#) and the use of photography as a tool for evidentiary classification in science (...including eugenics) and the arts. Examples: Alphonse Bertillon, Snowflake Bentley, Bernd and Hilla Becher, Stacey Green; as well as e.g. Karl Blossfeldt, Ed Ruscha, Nancy Burson, Jenny Odell (Satellite Collections), Dina Kelberman (I'm Google), Yufeng Zhao (AllTextInNYC), Hasan Elahi (Tracking Transience)



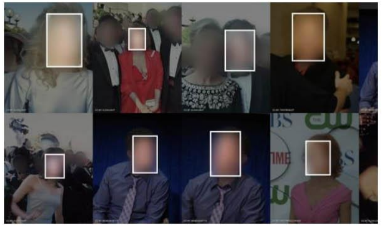


3.1c. Background: Confronting Dataset Bias with Exposing AI

- **Examine** Adam Harvey's [Exposing.ai](), "an art and research publication investigating the ethics, origins, and individual privacy implications of face recognition datasets created 'in the wild.'"
- **Write** a sentence of reflection on something you found interesting in this site.

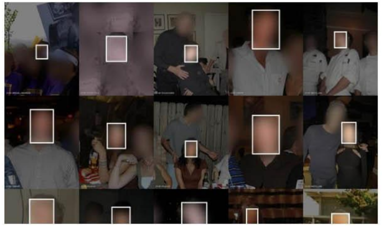
Duke Multi-Target Multi-Camera Tracking Dataset

Person re-identification, multi-camera tracking



FaceScrub

Face recognition and detection



FaceTracer

large database with variety of metadata and fiducial points marked and descriptive attributes

Flickr Diverse Faces Dataset

Ethnically diverse face recognition dataset

FFHQ Dataset

Face synthesis

GeoFaces

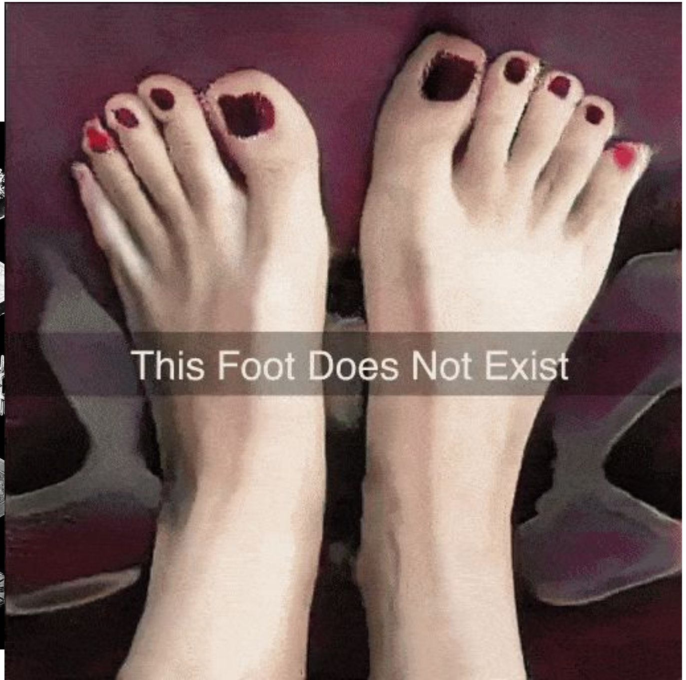
3D face recognition dataset

Students learn about public datasets like the DMTMC, in which Duke University published 2M+ surveillance images of its own students.

3.2. “More Like This, Please”

Students examine artworks in which artists have trained their own AI models in order to produce more like this” — e.g. Chrystal Y. Ding, Performance II: Skin on Repeat (2021); MSCHF, This Foot Does Not Exist (2019); Zach Whelan, AI Generates Comic Panels (2020).





3.3. Contemplate, Forage, Propose, Discuss

- **Read** *Making AI Art Responsibly: A Field Guide* by Emily Saltz, Lia Coleman, & Claire Leibowicz.
- **Search** and **browse** online for images, image datasets, and sources of images.
- **Propose** an example of a dataset you might work with for each of the following methods:
  - Your own **selfmade** image dataset(s) — your own photos, drawings, scans, etc.
  - Collection(s) of online media you'd be interested to assemble by **scraping**.
  - Existing ready-made datasets (or portions) that you'd like to **use**.
  (**Think** about: *Who* made this dataset? *Why*? Was the data ethically sourced?)

Summary of best practices:

- Least risky: Use your own illustrations, photography, text, or video.
- If you’re scraping work, prioritize work that is in the public domain, or directly ask for permission from those whose identity and/or work is represented in the dataset *(...see Fursona Controversy)*
- **Credit** the work of others whenever possible. If you’re posting online, tag people and thank them!
- If your dataset has people in it, ask yourself: How would I feel if my picture, or my friends and family’s pictures, were in this dataset? How can I ask for consent?

3.4. Bring in 500 Images

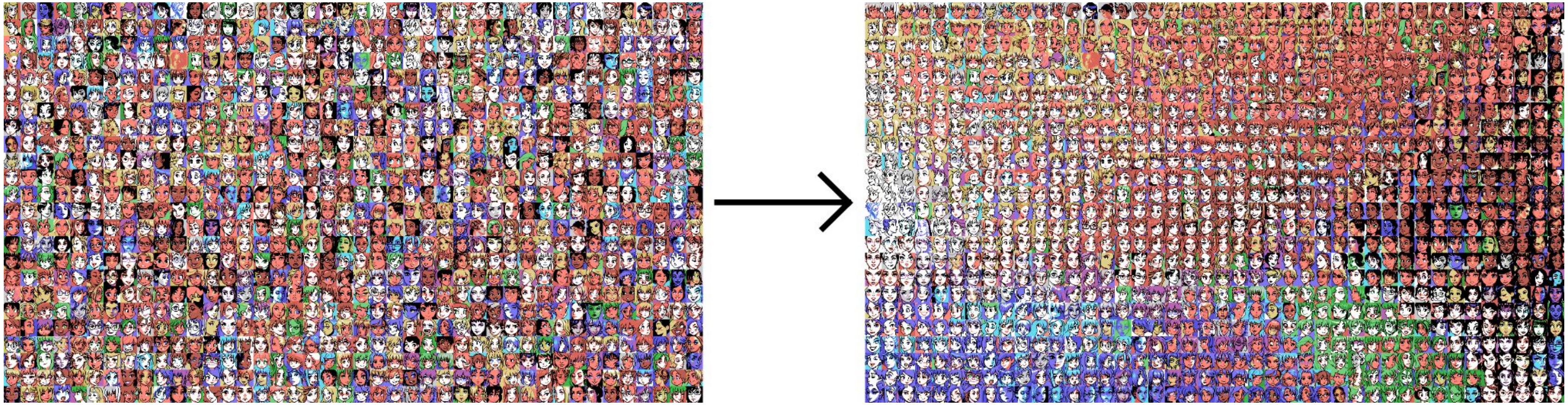
Students are asked to create or collect a dataset of 500 images. 500 is good — it's the scale at which avoiding the learning of a new automation tool (scrapy, ffmpeg) becomes difficult to justify.

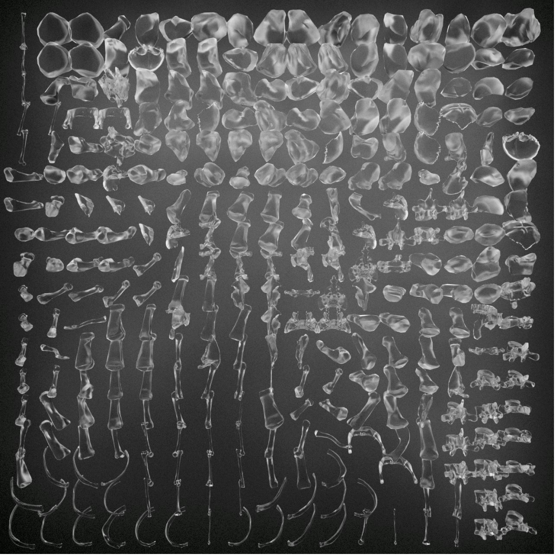
- **Create, find**, or **pilfer** a coherent collection of at least 500 images that intrigue you. You could make your own (from photography, video, or drawings); scrape existing media (using *Scrapy* or *BeautifulSoup*); or use an existing readymade dataset. Give credit and make good choices.
- Using `ffmpeg`, **create** directories containing new versions of your images, cropped and scaled to 32x32 and 128x128 pixels.
- **Compile** a simple image grid of all the 128x128 images, using the Imagemagick `montage` command. You may need to `brew install imagemagick`.
- **Post** your simple image grid to the class Discord.
- **Write** about your dataset. What did you collect? Why did this data intrigue you? How did you collect it? What challenges did you overcome to collect/organize/use it?

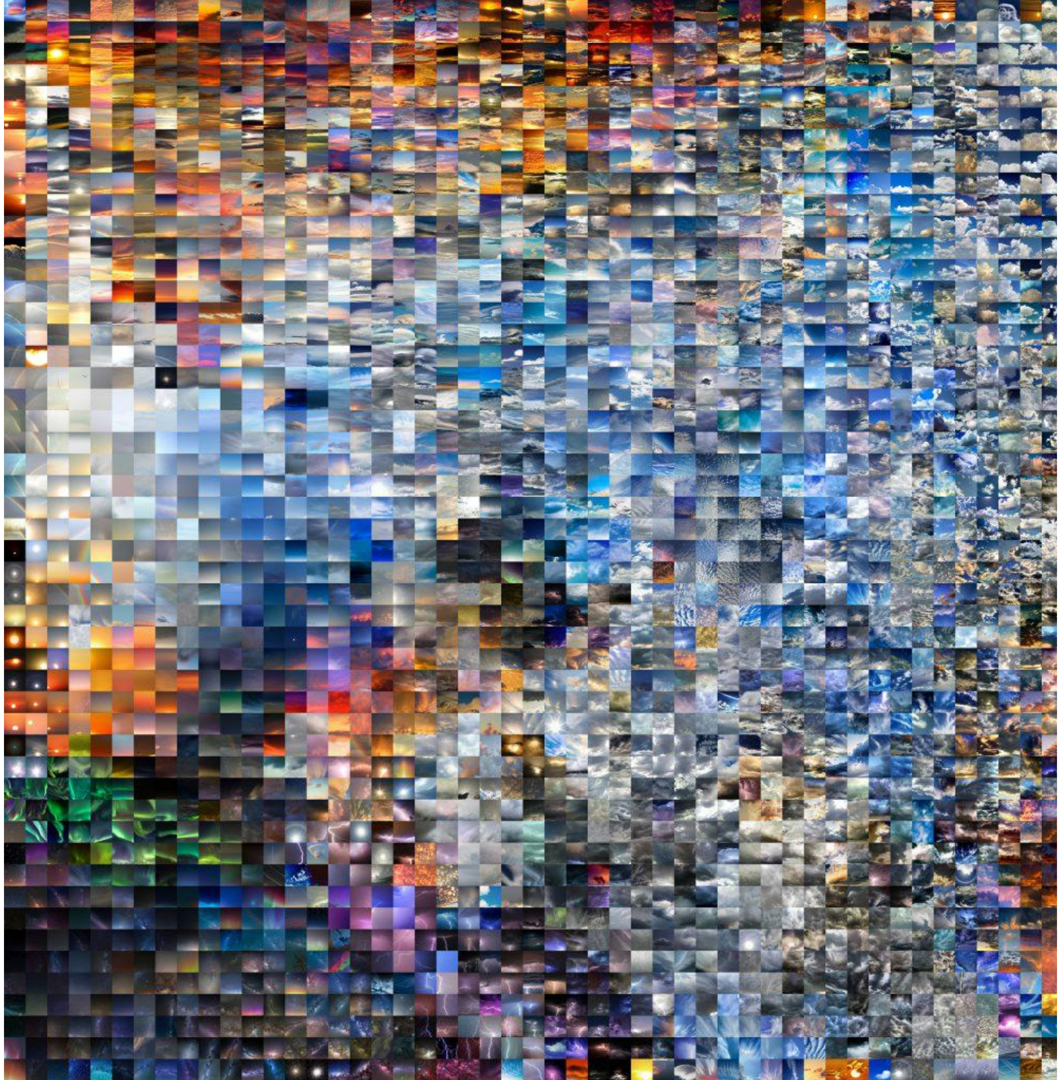
3.5. Clustering and Visualization: an ML-Assisted Mosaic

Students are provided with Kyle McDonald's ImageRearranger: a Jupyter/Colab Python notebook that produces a "rectified embedding grid visualization". It: analyzes the images with a CNN; performs dimensionality reduction and clustering (using UMAP or t-SNE); and solves the "Assignment Problem".

- **Use** the notebook to **make** a mosaic that presents your image dataset in an organized way.
- **Post** an image of your organized UMAP/t-SNE mosaic in the class Discord.
- **Write** about any discoveries, or aspects of your dataset which this process makes visible.



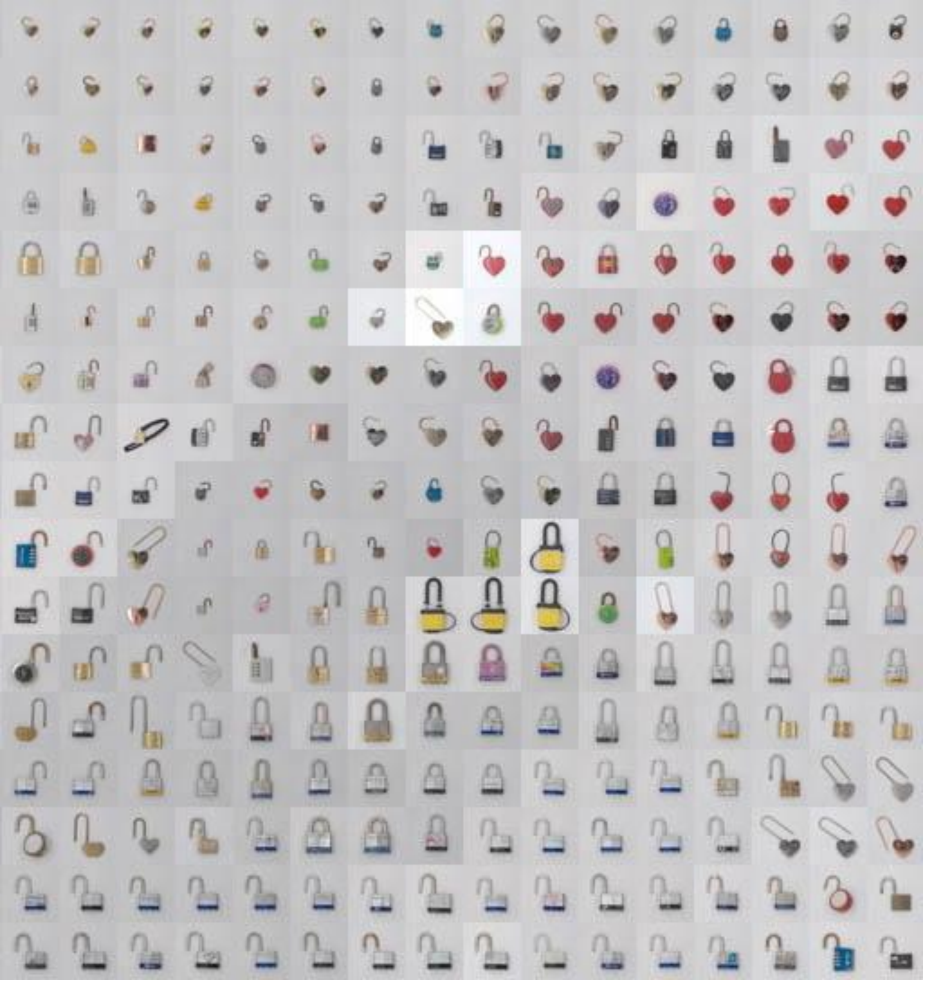




Zach Rispoli: *Ossuary* (BodyParts3D);

Christopher Wei: *EcstasyData.org*;

Kaitlin Schaer, *Skies worth Seeing*



Nicole Huang: *Love Locks from the Anderson Bridge*

Emmanuel Lugo: *Lower Teeth*

Jeffrey Wang: *All Legos*

Cathleen Zhang:
Portable Microscope Discoveries in One Square Meter of the Carpet at the CMU University Center

3.6. Cleaning Your Dataset

Now that they can see their entire dataset at a glance, students use some simple tools to remove outliers, prior to training their AI model…

- **Use** this Processing image collection **swiper** to organize images into keep-or-toss piles: https://github.com/golanlevin/gen-ai/blob/main/assignments/image_collection_swiper.zip
- **Use** this p5 umap-mouseover-**deleter** to remove images from your dataset: https://editor.p5js.org/golan/sketches/jlKwVFG4N

3.7. Train a Generative AI Model on Your Own Dataset

We will use Civitai.com to train a LoRA image model. (Caution: Civitai is a NSFW minefield.) We then use this model in ComfyUI to make "more like this, please".

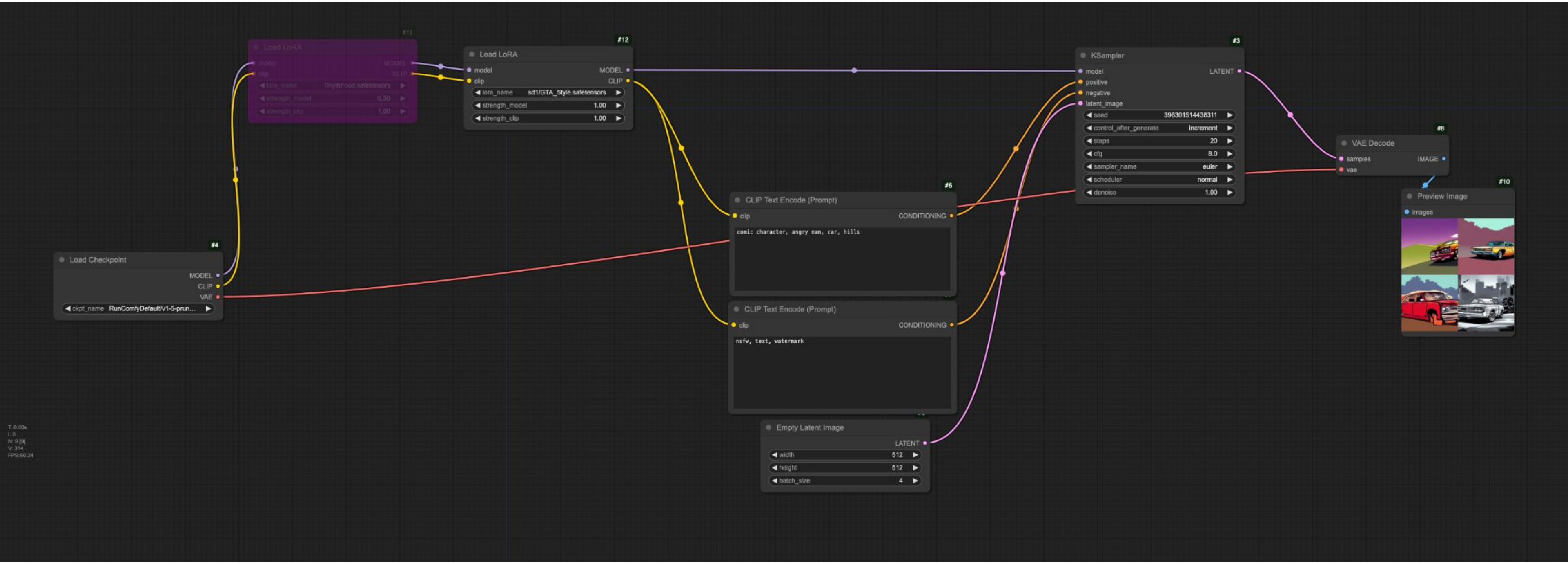
NOTE: Be prepared for the possibility that your dataset itself, as a never-before-seen object, or the act of collecting your dataset, as a kind of performance, might be more interesting than what you create with an AI trained on it.

- **Use** Civitai to train a LoRA on your images. (https://civitai.com/models/train)
- **Upload** your LoRA to RunComfy, and store it in the `models/lora` folder.
- **Connect** your LoRA in ComfyUI, and **use** it to generate some new images.

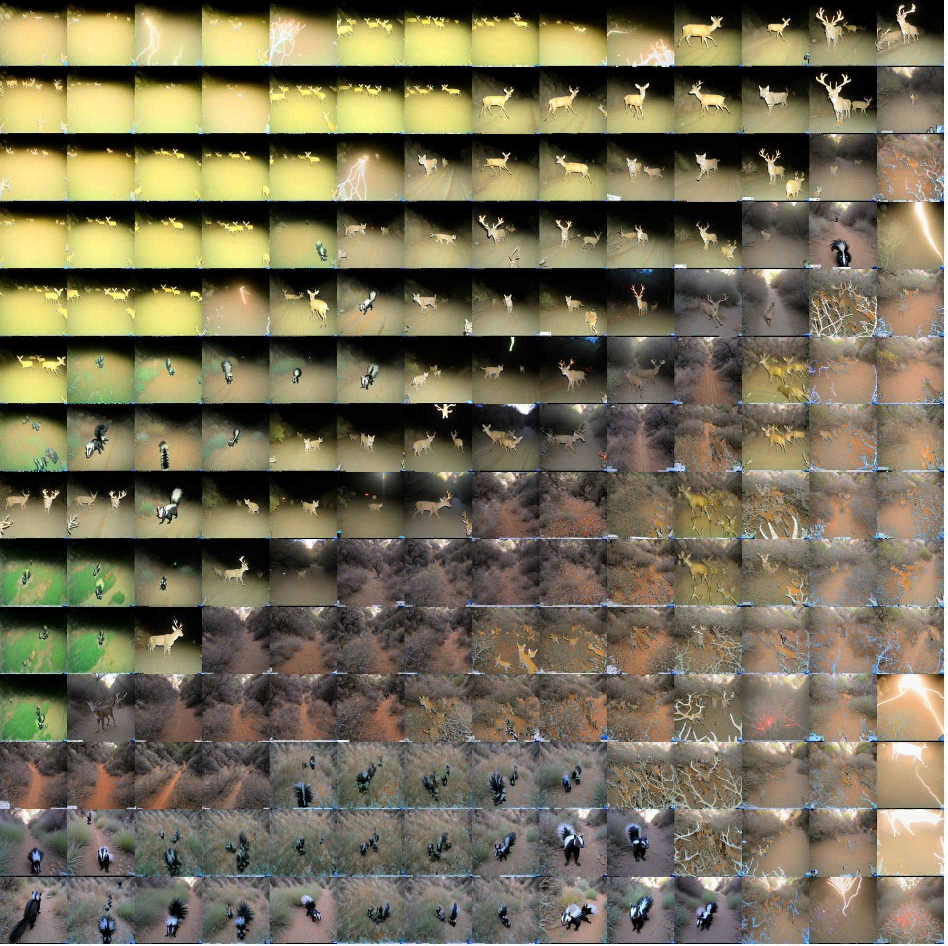
Images generated using a LoRA trained on *Les songes drolatiques de Pantagruel* (1565)

3.7. Example LoRA Workflow in ComfyUI

NOTE: *Be prepared for the possibility that your dataset itself (as a never-before-seen object) or the act of collecting your dataset (as a kind of performance), might be more interesting than what you create with an AI trained on it.*



https://github.com/golanlevin/gen-ai/blob/main/daily_notes/workflows/lora_example_workflow.png



A particularly successful response by Dario Quintero, a BCSA sophomore. Dario visualized hundreds of wildlife images that he acquired from a public online trailcam in California: https://github.com/DarioQuintero/caltech-trailcam-curation/, and ended up generating cryptids

Resources and Links

- The complete assignment:
  https://github.com/golanlevin/gen-ai/blob/main/assignments/assignment_3.md
- Daily notes and lectures for this unit:
  https://github.com/golanlevin/gen-ai/blob/main/daily_notes/0213.md
  https://github.com/golanlevin/gen-ai/blob/main/daily_notes/0218.md
  https://github.com/golanlevin/gen-ai/blob/main/daily_notes/0220.md
- Dario Quintero's project, *caltech-trailcam-curation*:
  https://github.com/DarioQuintero/caltech-trailcam-curation/

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